
IT623: Data Structures Academic Year 2025-2026 (Autumn)

Instructor:	Dr. Supantha Pandit[supantha.pandit@dau.ac.in]
Prerequisites:	Discrete Mathematics, C Programming
Slot/ Category:	M.Sc. IT, Semester I — Core Course
Course Credits:	4 Credits (3-0-2)
Foundation for:	CS/IT stream of courses
Lectures:	TBD
TA Contact Info:	TBD

Course Description

This course aims to introduce the concept of data structures, their indispensability in implementing algorithms, and how they aid in improving performance. Extensive coverage of the well-known and essential data structures such as arrays, linked lists, queues, stacks, hash tables, binary search trees, AVL trees, red-black trees, 2-3-4 trees, heaps, along with associated routines/algorithms such as dictionary operations, order statistics, sorting will constitute the contents of the course.

Suggested Books

- **Textbook:**

- *Data Structures and Algorithms* by Aho, Hopcroft, and Ullman, Addison-Wesley, 1999.

- **Reference Books:**

- *Data Structures and Algorithms in C++/Java* by Goodrich, Tamassia, and Goldberg, Wiley, 2011
- *Introduction to Algorithms* by Cormen, Leiserson, Rivest, and Stein, PHI, 2010

Course Outcomes

After successful completion of the course, the student will have the ability to:

- Utilize appropriate simple data structures in problem-solving
- Understand data abstraction and order statistics
- Understand program efficiency through analysis of algorithms

Program Outcomes Mapping:

P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
X	X	X		X				X			X

Evaluation Scheme¹

- **In-Semester Exams (I & II):** 40% (20% + 20%)
- **Course Project/ Lab Assignments:** 20%
- **End-Semester Exam:** 40%

Tentative Course Plan²

Unit	Topics	Number of Lectures
Preliminaries	Representation of data on a computer, data types; array and linked list representations; ways of representing programs and associated data on computers.	3
Analysis Tools	The notion of the running time of an algorithm, Recurrences, Parameters of performance.	3
Dictionary Operations	Find, Max, Min, Successor, Predecessor (query operations); Insert, Delete (modify operations).	6
List Data	Stacks, queues, variant implementations using arrays and linked lists, hashing.	6
Sorting	Comparison-based sorting algorithms, other sorting algorithms; lower bounds for comparison-based sorting algorithms; best-case, worst-case, and average-case running times; Heap Sort, Bubble Sort, etc.	9
Order Statistics	Maximum and minimum elements of a set, finding median, searching for an element of a given rank, finding the rank of a given element, ranks of a subset of elements, maintaining rank information for a dynamic set.	3
Trees	Heaps, Binary Search Trees (BSTs), height of BSTs, Tries, Balanced BSTs: Red-Black Trees, AVL Trees, B-Trees.	9
Graphs	Representation using adjacency matrices and adjacency lists; graph searching algorithms: BFS and DFS.	3

¹This evaluation scheme may slightly deviate and is subject to change without notice.

²This course plan is subject to change without notice.

Tentative Weekly Schedule

Week	Topics
Week 1	Course Admin; Introduction to Data Structures: Variables, Data Types, and Data Structures.
Week 2	Algorithm Analysis: Counting operations; Asymptotic Analysis; Best-, Worst-, and Average-case; Abstract Data Types (ADTs).
Week 3	Arrays: Definition, Types, Terminology, Computer Representation, Operations, and Applications.
Week 4	Linked Lists: Definition, Types, Terminology, Representation, Operations, and Applications.
Week 5	Order Statistics.
Week 6	Stacks: Definition, Types, Representation, Operations, and Applications.
Week 7	Queues: Definition, Types, Representation, Operations, and Applications.
Week 8	Tables: Definition, Types, Representation, Operations, and Applications.
Week 9	Trees: Binary Trees and its Types; Implementation in Arrays; Traversals: Inorder, Preorder, Postorder; Basic Operations.
Week 10	BST: Traversals and Sorting; Insertion, Deletion, Augmented BSTs and Update Costs.
Week 11	Class rescheduling window.
Week 12	AVL Trees: Min and Max Heights, Searching, Insertion.
Week 13	Heaps: Insertion, Deletion, Heapify, HeapSort, Priority Queues.
Week 14	Graphs: Representations using Adjacency Matrix/List; BFS and DFS Algorithms.